Inflation relative price variance

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1 Introduction

Inflation is known to have adverse welfare effects through its distortion of relative prices which causes the misallocation of resources, production inefficiencies leading to lower real economic activity. This is well supported by both theoretical and empirical studies. This paper contributes to the empirical literature, by examining the relationship between inflation and relative price variability (RPV) in South Africa given the little evidence available and to better understand how this relationship has evolved and the implications for monetary policy.

The South African context can provide further insights into the relationship between inflation and RPV given the shifts in the inflation targets from 3- 6 per cent in 2000 to 4.5 per cent since 2017 as structural changes in inflation can affect the link between inflation and relative price variability (Caglayan and Filiztekin, 2003) and that the functional form of relationship is U-shaped and dynamic (Fielding and Mizen, 2000; Choi, 2010). The latter motivates the estimation of the inflation rate that minimizes relative price distortions since if the relationship was linear then the optimal inflation rate would be zero. Given the inflation target reform towards a 3% in South Africa, this study contributes to the policy discussion on the appropriate inflation target and understating how dynamic relations between inflation and relative price variability has evolved.

The theoretical literature generally distinguishes between menu-cost and signal extraction models in explaining the positive relation between trend inflation and RVP. In these models' firms follow a one-sided dynamic pricing rule when setting prices and face fixed non-zero costs of adjusting nominal prices (e.g. printing new menus) when inflation increases as the latter reduces the optimal real price of goods and services. Due to differences in administrative costs, these nominal price adjustment costs vary across firms moreover these adjustments occur at different times which implies staggard price setting behaviour which leads to relative price variability. Menu cost models predict a that higher expected inflation causes an increase in RVP (citations xxx) while Lucas-type signal extraction models (Lucas(1973); Hercowitz (1981); Cukierman (1984) xxxx more citatiosn) argue unexpected inflation uncertainty causes higher RVP because informational frictions make it difficult to distinguish between idiosyncratic relative shocks or an aggregate (inflationary) shock. Menu cost and signal-extraction models assume a positive and linear relationship making the optimal inflation rate zero, however, recently monetary search models (Head and Kumar, 2005) show that this relationship is nonlinear making the optimal inflation rate somewhere between zero and positive inflation rate.

2 Citations

Adam et al. (2024) Baglan et al. (2016) Ball and Romer (1993) Benabou and Gertner (1993) Caglayan and Filiztekin (2003) Caglayan et al. (2008) Grier and Perry (1996) Lach and Tsiddon (1992) Nakamura et al. (2018) Ndou and Redford (2014) Ndou and Gumata (2017) Parks (1978) Parsley (1996) Rather et al. (2018)

3 Literature review

4 Data and methodology

Our analysis uses monthly inflation data on 45 goods categories from January 2009 to March 2025 obtained from Statistics South Africa (Stats SA).[1] **The data is seasonally adjusted using ???.** Figure 1 below shows the long term inflation rate in South Africa, and Figure A1 in the appendix shows the various price categories used in the analysis. Table A1 provides the descriptive statistics of the inflation rates for the various price categories used in the analysis.

[1]http://www.statssa.gov.za/

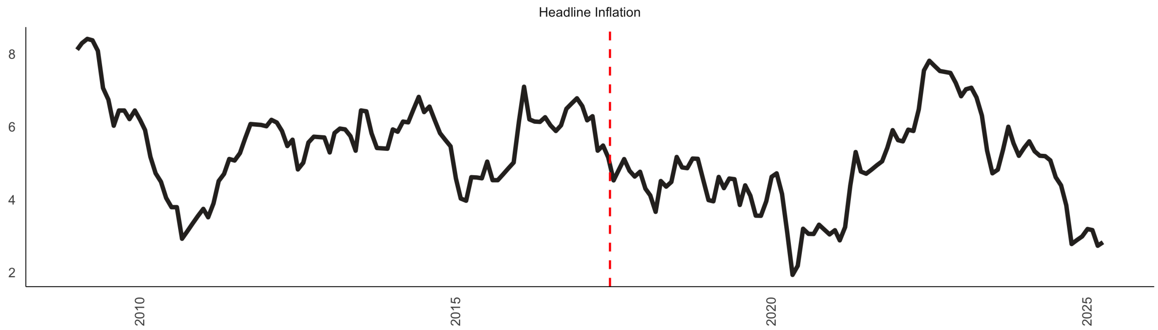

Figure 1: Headline inflation. Note: The dashed red line represents the implementation of the mid-point target of 4.5% in 2017.

The relative price variance (RPV) is calculated in line with Lach and Tsiddon (1992) as the weighted standard deviation of individual price inflation rates from the aggregate inflation rate. The RPV is essentially the variability of relative prices of a given product. We first calculate the monthly inflation rate as follows:

$$\pi_{i,t} = ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right) * 100, \tag{1}$$

where $\pi_{i,t}$ is the inflation rate of good i at time t, $P_{i,t}$ is the price of good i at time t, and $P_{i,t-1}$ is the price of good i at time $t-1$. Then, the RPV is calculated as:

$$RPV_t = \sqrt{\sum_{i=1}^{n} \omega_i (\pi_{i,t} - \pi_t)^2}, \tag{2}$$

where ω_j is the expenditure share of the j^{th} good, π_t is the headline or aggregate inflation rate at time t, and $\sum_{i=1}^{n} \omega_i = 1$. The expenditure weights (ω_i) used are derived from the Consumer Price Index (CPI) basket provided by Stats SA. Figure 2 shows the RPV calculated using the above formula.**When where these weigths constructed???**. The weights are shown in Figure A2 in the appendix.

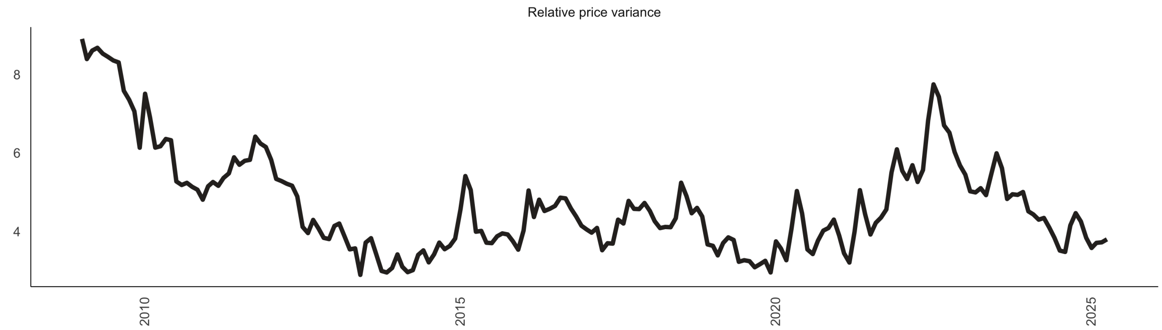

Figure 2: Relative price variance

To examine the relationship between inflation and RPV, we estimate the following econometric model:

$$RPV_t = \beta_0 + \beta_1, |\pi_t| + \sum_{h=1}^{4} \omega_h RVP_{t-h} + \mu_t, \tag{3}$$

where RPV_t is the relative price variance at time t, $|\pi_t|$ is the absolute value of the inflation rate at time t, RVP_{t-h} are lagged values of RPV to account for its dynamic nature, and μ_t is the error term. The coefficient β_1 captures the effect of inflation on RPV. We include up to four lags of RPV based on Akaike Information Criterion (AIC) and to capture potential persistence in relative price variability.

The literature on RVP emphasises the that the relationship between inflation and RPV may be time-varying due to structural changes in the economy, monetary policy regimes, and other factors. To capture potential time variation in the relationship between inflation and RPV, we also estimate a rolling regression model with a **24-month and 60-month windows** as follows:

$$RPV_t = \alpha_t + \beta_{1,t}|\pi_t| + \sum_{h=1}^{4} \omega_{h,t} RVP_{t-h} + \mu_t, \tag{4}$$

where the coefficients α_t, $\beta_{1,t}$, and $\omega_{h,t}$ are allowed to vary over time. This approach allows us to observe the stability of the coefficients over time, which is an indicator of non-linearities in the relationship between RPV and inflation. Furthermore, we employ a quantile regression approach, which is robust to outliers and allows us to examine the relationship between inflation and RPV across different points of the RPV distribution. The quantile regression model is specified as follows:

$$Q_\tau(RPV_t|\pi_t, RPV_{t-1}, ..., RPV_{t-4}) = \gamma_0 + \gamma_1|\pi_t| + \sum_{h=1}^{4} \delta_h RVP_{t-h}, \tag{5}$$

where $Q_\tau(RPV_t|\pi_t, RPV_{t-1}, ..., RPV_{t-4})$ represents the τ^{th} quantile of RPV conditional on inflation and its lagged values. The coefficients γ_0, γ_1, and δ_h capture the effects at different quantiles of the RPV distribution.

Lastly, we utilise a semi-parametric or partially linear approach to estimate the optimal level of inflation given the RVP. This approach allows us to model the relationship between inflation and RPV without imposing a strict functional form. Following the literature, we first specify a quadratic (U-shaped) relationship between inflation and RPV as follows:

$$RVP_t = \alpha + \pi_t + \beta_2\pi_t^2 + \sum_{h=1}^{4} \phi_h RVP_{t-h} + \epsilon_t, \tag{6}$$

where β_2 captures the curvature of the relationship between inflation and RPV. The optimal inflation rate that minimizes RPV can be derived by taking the first derivative of the equation with respect to $|\pi_t|$ and setting it to zero, yielding:

$$\pi_t^* = -\frac{1}{2\beta_2}. \tag{7}$$

To estimate this optional inflation we implement a semi parametric approach as follows:

$$RVP_t = f(\pi_t) + \sum_{h=1}^{4} \phi_h RVP_{t-h} + \epsilon_t, \tag{8}$$

where $f(\pi_t)$ is a non-linear smooth differentiable function of inflation and the RVP. Therefore, the optimal inflation rate can be obtained by finding the minimum of the estimated $f(\pi_t)$ function. $f(\pi_t)$ is estimated in two stages. First we estimate λ_h as follows:

$$RVP_t = \sum_{h=1}^{4} \lambda_h \overline{RVP}_{t-h} + \eta_t, \tag{9}$$

where $\overline{RVP}_{t-h}$ are the residuals from a non-parametric regression of each lag of RVP_t on π_t . In the second stage, we estimate $f(\pi_t)$ non-parametrically as follows:

$$\hat{\eta}_t = f(\pi_t) + v_t, \tag{10}$$

where $\hat{\eta}_t$ are the residuals from Equation 9, and $f(\pi_t)$ is estimated using a kernel regression method. In both stages we use the Nadaraya-Watson kernel estimator. The bandwidth (h) for the kernel estimator is selected using **cross-validation to minimize the mean squared error**. As a treatment for extreme values of inflation, we further implement the unbounded Gaussian kernel and the outlier-robust Epanechnikov kernel. In essence, the semi-parametric approach captures the sensitivity of RPV to marginal increases in inflation. In that case, the optimal inflation rate can be obtained by finding the point where the first derivative of $f(\pi_t)$ ($f'(\pi_t)$) is equal to zero. Therefore, if $f'(\pi_t) > 0$ ($f'(\pi_t) < 0$), then an increase in inflation (decrease) leads to an increase (decrease) in RPV.

5 Results

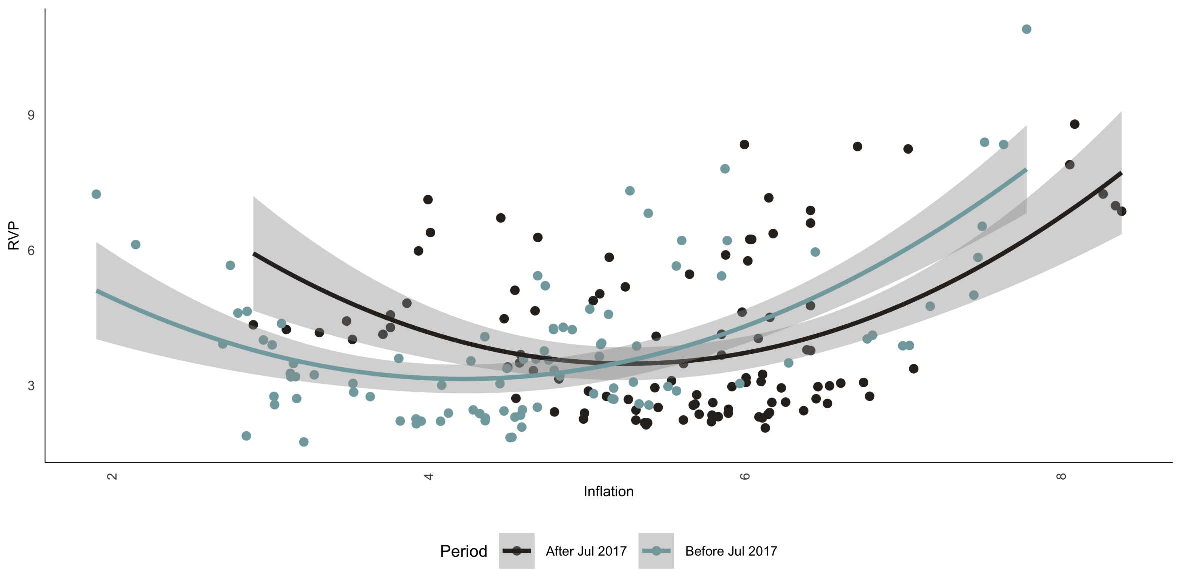

Figure 3: Relative price dispersion vs Inflation

Table 1: Baseline regression results of RPV on absolute inflation

	Estimate
RVP Full Sample	
(Intercept)	0.15*
headline_inflation	0.01
lag(log_rvp, 1)	1.07***
lag(log_rvp, 2)	-0.53***
lag(log_rvp, 3)	0.31***
RVP Pre 2017	
(Intercept)	0.06
headline_inflation	0.01
lag(log_rvp, 1)	0.95***
lag(log_rvp, 2)	-0.19
lag(log_rvp, 3)	0.14
RVP Post 2017	
(Intercept)	0.19*
headline_inflation	0.02
lag(log_rvp, 1)	1.13***
lag(log_rvp, 2)	-0.72***
lag(log_rvp, 3)	0.38***
RVP Full Sample with Squared Inflation	
(Intercept)	-0.28
headline_inflation	0.46*
squared_log_headline_infaltion	-0.71
lag(log_rvp, 1)	1.05***
lag(log_rvp, 2)	-0.51***
lag(log_rvp, 3)	0.29***
RVP Pre 2017 with Squared Inflation	
(Intercept)	-0.46*
headline_inflation	0.63**
squared_log_headline_infaltion	-0.99**
lag(log_rvp, 1)	0.92***
lag(log_rvp, 2)	-0.16
lag(log_rvp, 3)	0.11
RVP Post 2017 with Squared Inflation	
(Intercept)	-0.43
headline_inflation	0.68
squared_log_headline_infaltion	-1.01
lag(log_rvp, 1)	1.06***
lag(log_rvp, 2)	-0.69***
lag(log_rvp, 3)	0.33***

Note: *** = p value < 0.01, ** = 0.01 < p value < 0.05, * = 0.05 < p value < 0.1.

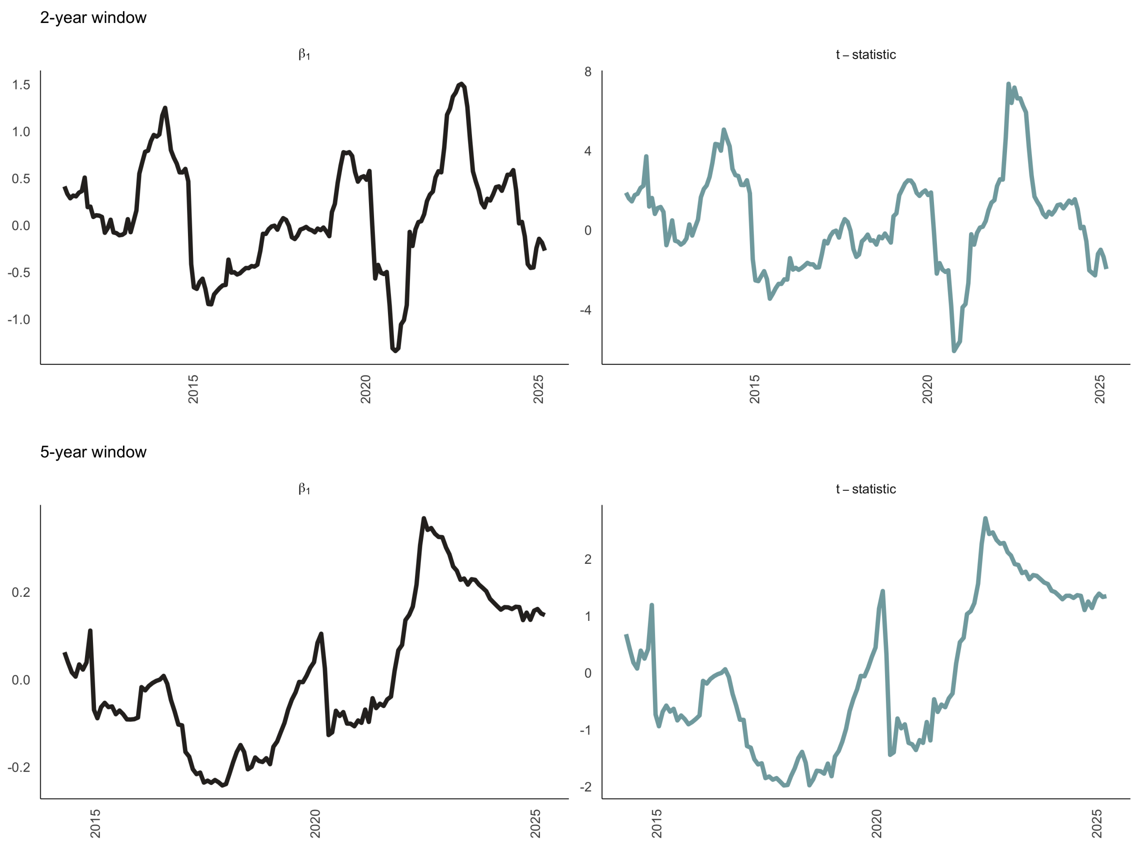

Figure 4: Rolling regression coefficients (2 year and 5 year windows)

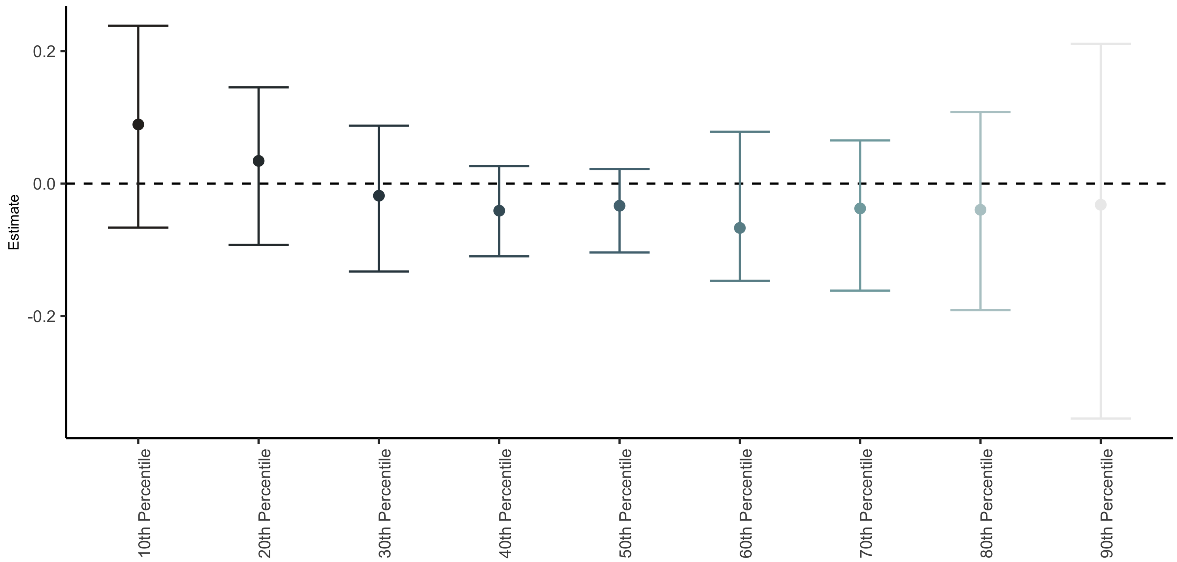

Figure 5: RVP quantile regressions estimates

6 Conclusion

References

Adam, K., Alexandrov, A., and Weber, H. (2024). *Inflation Distorts Relative Prices: Theory and Evidence*. Number 23/2024. Deutsche Bundesbank Discussion Paper.

Baglan, D., Yazgan, M. E., and Yilmazkuday, H. (2016). Relative price variability and inflation: New evidence. *Journal of Macroeconomics*, 48:263–282.

Ball, L. M. and Romer, D. H. (1993). Inflation and the Informativeness of Prices.

Benabou, R. and Gertner, R. (1993). Search with learning from prices: Does increased inflationary uncertainty lead to higher markups? *The Review of Economic Studies*, 60(1):69–93.

Caglayan, M. and Filiztekin, A. (2003). Nonlinear impact of inflation on relative price variability. *Economics Letters*, 79(2):213–218.

Caglayan, M., Filiztekin, A., and Rauh, M. T. (2008). Inflation, price dispersion, and market structure. *European Economic Review*, 52(7):1187–1208.

Choi, C.-Y. (2010). Reconsidering the Relationship between Inflation and Relative Price Variability. *Journal of Money, Credit and Banking*, 42(5):769–798.

Fielding, D. and Mizen, P. (2000). Relative Price Variability and Inflation in Europe. *Economica*, 67(265):57–78.

Grier, K. B. and Perry, M. J. (1996). Inflation, inflation uncertainty, and relative price dispersion: Evidence from bivariate GARCH-M models. *Journal of Monetary Economics*, 38(2):391–405.

Head, A. and Kumar, A. (2005). Price Dispersion, Inflation, and Welfare. *International Economic Review*, 46(2):533–572.

Lach, S. and Tsiddon, D. (1992). The Behavior of Prices and Inflation: An Empirical Analysis of Disaggregat Price Data. *Journal of Political Economy*, 100(2):349–389.

Nakamura, E., Steinsson, J., Sun, P., and Villar, D. (2018). The elusive costs of inflation: Price dispersion during the US great inflation. *The Quarterly Journal of Economics*, 133(4):1933–1980.

Ndou, E. and Gumata, N. (2017). *Inflation Dynamics in South Africa: The Role of Thresholds, Exchange Rate Pass-through and Inflation Expectations on Policy Trade-Offs*. Springer.

Ndou, E. and Redford, S. (2014). Relative price variability: Which components of the consumer price index contribute towards its variability? *South African Reserve Bank Working Paper Series*.

Parks, R. W. (1978). Inflation and Relative Price Variability. *Journal of Political Economy*, 86(1):79–95.

Parsley, D. C. (1996). Inflation and relative price variability in the short and long run: New evidence from the United States. *Journal of Money, Credit and Banking*, 28(3):323–341.

Rather, S. R., Durai, R. S., and Ramachandran, M. (2018). Inflation and the Dispersion of Relative Prices: A Case for 4 % Solution. *Australian Economic Papers*, 57(1):81–91.

A Appendix A: Data and descriptive statistics

A.1 Data

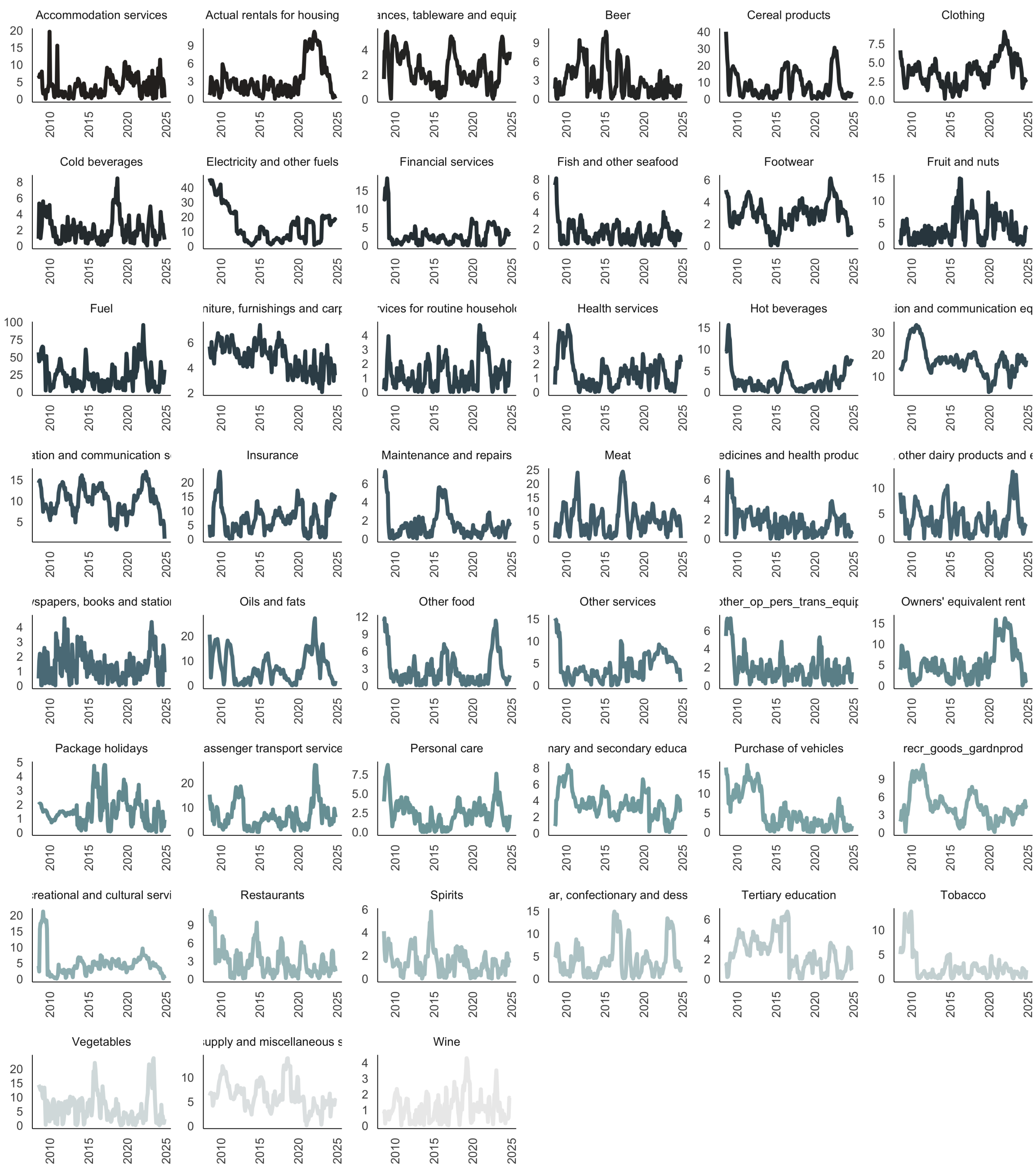

Figure A1: Good categories inflation

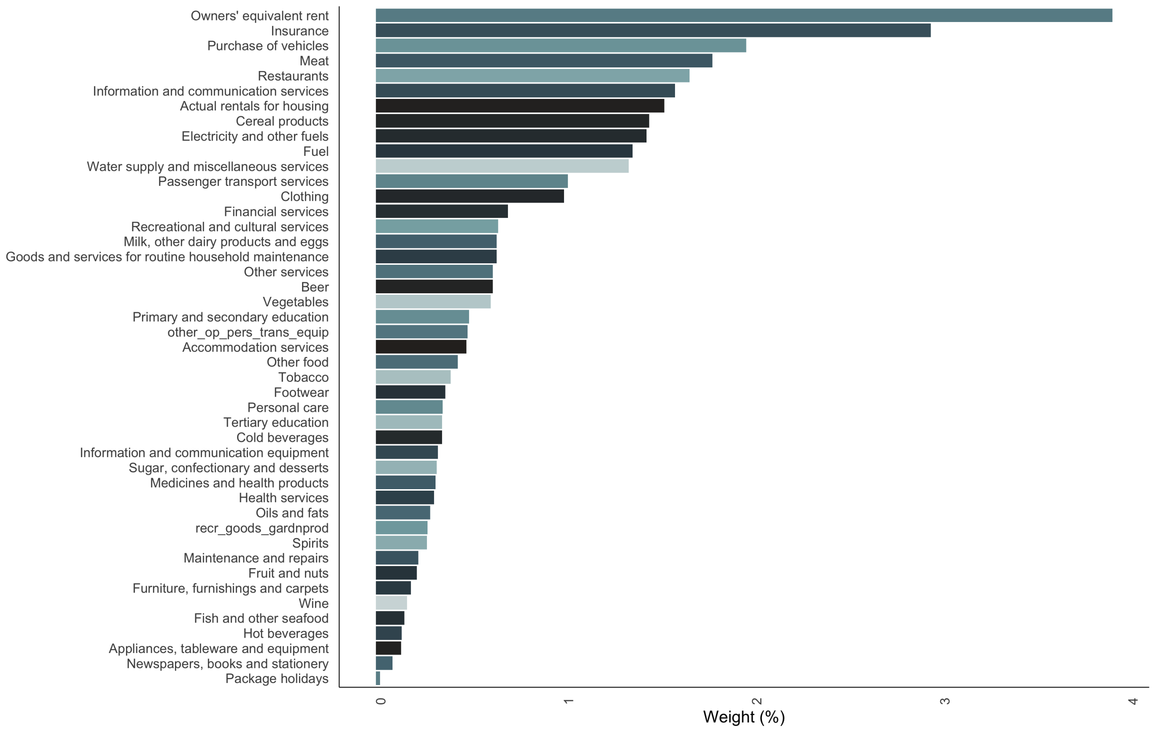

Figure A2: CPI Weights. Note: Due to a lack of data, some categories (no spitirits; transport services of goods; household textiles; glassware,tableware, and house utensils; and Tools and equipment for house and garden) have been excluded.

A.2 Descriptive statistics

Table A1: Descriptive statistics

Goods category	Mean	Median	SD	Min	Max	Observations
Accommodation services	3.69	2.97	2.96	0.02	19.60	196
Actual rentals for housing	3.19	2.39	2.73	0.04	11.28	196
Appliances, tableware and equipment	2.40	2.13	1.35	0.03	5.47	196
Beer	2.97	2.04	2.53	0.01	10.77	196
Cereal products	8.33	4.98	7.98	0.01	40.01	196
Clothing	3.74	3.61	1.78	0.17	9.09	196
Cold beverages	2.01	1.62	1.57	0.04	8.45	196
Electricity and other fuels	13.98	10.03	11.77	0.53	46.31	196
Financial services	2.76	1.97	3.14	0.02	18.30	196
Fish and other seafood	1.46	1.15	1.24	0.01	8.09	196
Footwear	2.84	2.87	1.22	0.03	6.13	196
Fruit and nuts	3.69	2.99	3.07	0.02	14.90	196
Fuel	22.89	18.12	18.11	0.13	95.60	196
Furniture, furnishings and carpets	4.77	4.80	1.08	2.07	7.39	196
Goods and services for routine household maintenance	1.25	0.97	1.05	0.00	4.71	196
Health services	1.37	1.24	1.10	0.00	4.75	196
Hot beverages	2.74	1.94	2.75	0.03	15.61	196
Information and communication equipment	17.26	16.87	6.42	2.86	33.39	196
Information and communication services	10.04	10.12	3.25	0.91	16.99	196
Insurance	7.27	7.10	4.85	0.12	23.75	196
Maintenance and repairs	1.51	1.09	1.53	0.01	7.29	196
Meat	7.12	6.30	5.30	0.15	24.32	196
Medicines and health products	1.72	1.54	1.26	0.01	6.79	196
Milk, other dairy products and eggs	3.99	3.64	2.81	0.05	13.17	196
Newspapers, books and stationery	1.31	1.11	1.00	0.02	4.59	196
Oils and fats	7.39	6.15	5.99	0.00	27.17	196
Other food	2.67	1.70	2.74	0.06	11.73	196
Other services	4.15	3.19	3.07	0.00	15.10	196
Owners' equivalent rent	5.29	4.75	3.98	0.01	16.18	196
Package holidays	1.55	1.40	1.00	0.04	4.80	196
Passenger transport services	6.74	5.40	5.36	0.00	27.30	196
Personal care	2.44	2.33	1.77	0.01	8.72	196
Primary and secondary education	3.60	3.39	1.69	0.13	8.37	196
Purchase of vehicles	5.22	3.33	4.67	0.08	17.30	196
Recreational and cultural services	4.75	4.05	3.89	0.07	21.16	196
Restaurants	2.91	2.29	2.42	0.07	11.17	196
Spirits	1.43	1.30	1.04	0.00	5.80	196
Sugar, confectionary and desserts	4.18	3.18	3.75	0.03	14.87	196
Tertiary education	2.61	2.50	1.71	0.02	6.78	196
Tobacco	2.41	1.57	2.89	0.01	13.75	196
Vegetables	6.35	5.05	5.12	0.00	23.81	196
Water supply and miscellaneous services	6.16	5.77	2.83	0.15	13.91	196
Wine	1.13	0.98	0.83	0.03	4.28	196
other_op_pers_trans_equip	1.85	1.44	1.58	0.00	7.33	196
recr_goods_gardnprod	4.39	3.86	2.56	0.06	11.34	196

A.3 Stationarity tests

Table A2: Stationarity tests results

Measure	ADF test (p-value)	PP test (p-value)	KPSS test (p-value)	Stationarity
headline_inflation	0.19	0.24	0.08	Stationary
rvp	0.20	0.01	0.03	Stationary
rvp_division	0.22	0.01	0.03	Stationary

A.4 Lag length selection

A Appendix B: Alternative Results

A.1 Coefficient of variation version of RPV

The coefficient of variation (CV) version of RPV (or RVP-CV) normalizes the RPV by the aggregate price level to account for changes in the overall price level. This measure is particularly useful when comparing RPV across different time periods or economies with varying inflation rates. The RVP-CV is calculated as follows:

$$RPV_t = \sqrt{\frac{\sum_{i=1}^{n} \omega_i (\pi_{i,t} - \pi_t)^2}{|1 + \pi_t|}}, \tag{11}$$

where π_t is the aggregate inflation rate at time t, and ω_i are the inflation basket weights.

A.2 RPV-CV over-time

Figure B1 shows the CV version of RPV calculated using the above formula.

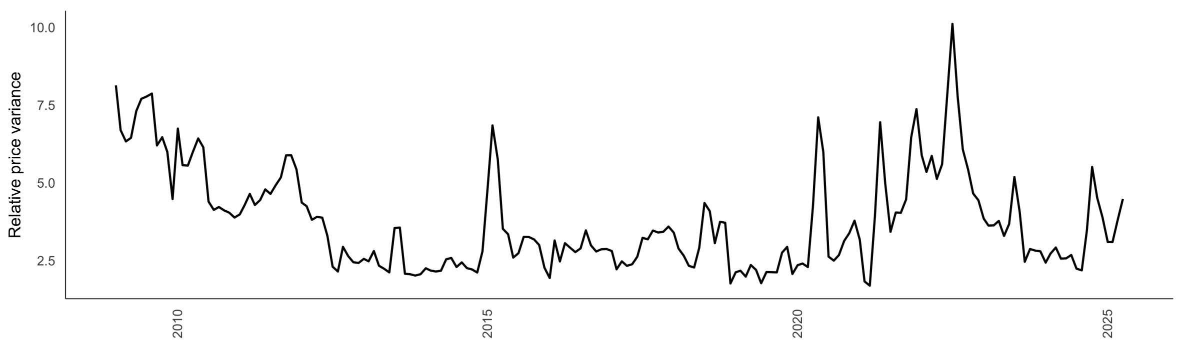

Figure B1: RVP-CV over-time